

WIPER AND RAIN SENSING

High-Level Design Document

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Classification: Synthetic Demonstration Data

1. System Overview

The Wiper and Rain Sensing coordinates its functions through typed AUTOSAR interfaces and ports. Software components exchange signals through provider and consumer ports.

2. Software Architecture

The primary software components are WiperManager, RainSensorControl, WiperMotorControl, WasherControl, StalkSwitchManager, and RearWiperControl.

3. Software Components

Component	Role	Primary Responsibility
WiperManager	Coordinator	Selects wiper mode from sensor and switch inputs.
RainSensorControl	Provider	Measures rain intensity.
WiperMotorControl	Consumer	Drives the front wiper motor.
WasherControl	Consumer	Controls the washer pump.
StalkSwitchManager	Provider	Reads wiper stalk switch.
RearWiperControl	Consumer	Drives the rear wiper.

4. Interfaces

Interface	Provider	Consumer	Signals
IRainIntensity	RainSensorControl	WiperManager	RainIntensity
IStalkInput	StalkSwitchManager	WiperManager	StalkSetting
IWiperDrive	WiperManager	WiperMotorControl	WipeInterval
IWasherDrive	WiperManager	WasherControl	WasherPulse
IRearWiperDrive	WiperManager	RearWiperControl	RearWipeMode

5. Ports

Component	Port	Direction	Interface
RainSensorControl	RainIntensity_Out	P-PORT	IRainIntensity
WiperManager	RainIntensity_In	R-PORT	IRainIntensity
StalkSwitchManager	StalkInput_Out	P-PORT	IStalkInput
WiperManager	StalkInput_In	R-PORT	IStalkInput
WiperManager	WiperDrive_Out	P-PORT	IWiperDrive
WiperMotorControl	WiperDrive_In	R-PORT	IWiperDrive
WiperManager	WasherDrive_Out	P-PORT	IWasherDrive
WasherControl	WasherDrive_In	R-PORT	IWasherDrive
WiperManager	RearWiperDrive_Out	P-PORT	IRearWiperDrive
RearWiperControl	RearWiperDrive_In	R-PORT	IRearWiperDrive

6. Signals

Signal	Data Type	Source	Destination
RainIntensity	uint8	RainSensorControl	WiperManager
StalkSetting	uint8	StalkSwitchManager	WiperManager
WipeInterval	uint16	WiperManager	WiperMotorControl
WasherPulse	uint8	WiperManager	WasherControl
RearWipeMode	uint8	WiperManager	RearWiperControl

7. Dependencies

Source	Relationship	Target	Reason
RainSensorControl	PROVIDES_INTERFACE	IRainIntensity	Publishes rain intensity.
WiperManager	REQUIRES_INTERFACE	IRainIntensity	Consumes rain intensity.
StalkSwitchManager	PROVIDES_INTERFACE	IStalkInput	Publishes stalk setting.
WiperManager	REQUIRES_INTERFACE	IStalkInput	Consumes stalk setting.
WiperManager	PROVIDES_INTERFACE	IWiperDrive	Publishes wipe interval.
WiperMotorControl	REQUIRES_INTERFACE	IWiperDrive	Consumes wipe interval.
WiperManager	PROVIDES_INTERFACE	IWasherDrive	Publishes washer pulse.
WasherControl	REQUIRES_INTERFACE	IWasherDrive	Consumes washer pulse.
WiperManager	PROVIDES_INTERFACE	IRearWiperDrive	Publishes rear wipe mode.
RearWiperControl	REQUIRES_INTERFACE	IRearWiperDrive	Consumes rear wipe mode.

8. Functional Flows

Flow F-001: Rain Intensity Distribution

RainSensorControl publishes RainIntensity through IRainIntensity. WiperManager receives the signal through RainIntensity_In and updates its internal state model.

Flow F-002: Stalk Setting Distribution

StalkSwitchManager publishes StalkSetting through IStalkInput. WiperManager receives the signal through StalkInput_In and updates its internal state model.

Flow F-003: Wipe Interval Distribution

WiperManager publishes WipeInterval through IWiperDrive. WiperMotorControl receives the signal through WiperDrive_In and updates its internal state model.

Flow F-004: Washer Pulse Distribution

WiperManager publishes WasherPulse through IWasherDrive. WasherControl receives the signal through WasherDrive_In and updates its internal state model.

Flow F-005: Rear Wipe Mode Distribution

WiperManager publishes RearWipeMode through IRearWiperDrive. RearWiperControl receives the signal through RearWiperDrive_In and updates its internal state model.

9. Integration Constraints

All provider and consumer ports shall reference compatible interfaces. Signal names shall remain consistent between interface definitions and functional flows. Architecture dependencies must be traceable to an approved source section.

10. Engineering Traceability

This synthetic HLD supports extraction, retrieval, validation and revision comparison experiments.